Simplified model of the NiC2 planes in LaNiC2

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Contents

1 Model **2**

2 Bandstructure **4**

3 Finite size results: ubiquitous topological modes **6**

1 Model

Jorge described a simplified model of the NiC_2 planes in $LaNiC_2$, which is depicted in Figure 1. In k-space the model is described by the following tightbinding Hamiltonian

$$\mathcal{H} = -\mu \mathbb{1}_2 + \mathcal{T}, \tag{1}$$

where μ is the chemical potential, $\mathbb{1}_2$ is the 2×2 identity matrix and $\mathcal{T}$ is the hopping matrix

$$\mathcal{T} = \alpha\sigma_+ + \beta\sigma_- + \text{h.c.}, \tag{2}$$

$\sigma_\pm$ are the Pauli raising and lowering matrices, α and β parameterise the hopping

$$\alpha = -v - we^{-ik_x} - 2t_d \cos(k_y) e^{-ik_x}, \tag{3}$$

$$\beta = -2t_d \cos(k_y), \tag{4}$$

where v is the intradimer hopping, w is the interdimer hopping and t_d is a diagonal hopping which stacks the SSH chains. In particular, note that the SSH model is topological only when $w > v$, however, in the model Jorge describes, $v > w$.

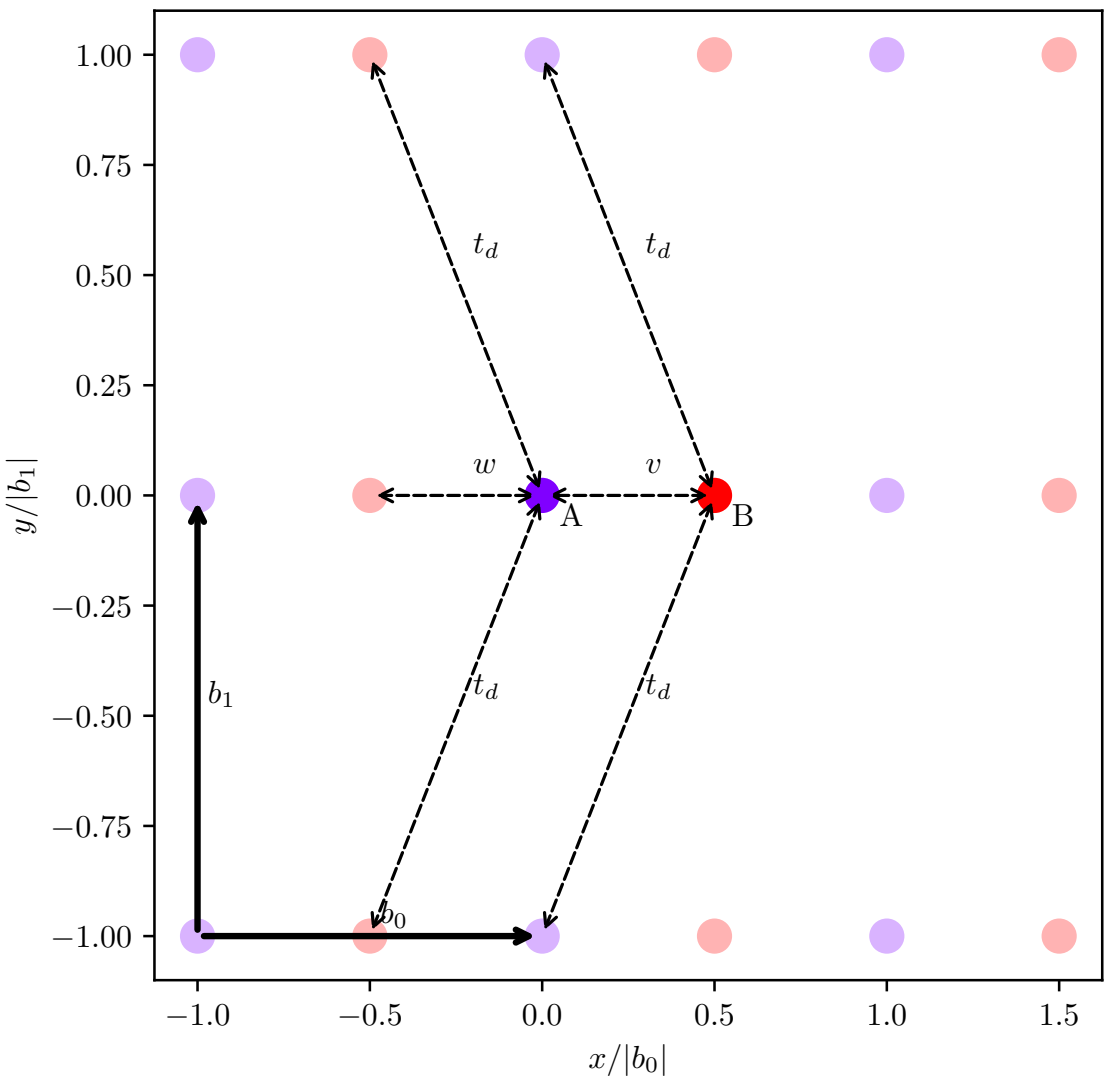

Figure 1: Planar model of the NiC_2 planes of $LaNiC_2$. Along the $\hat{x}$-direction, atomic sites A and B within the unit cell form an SSH chain, with intracell hopping v and intercell hopping w. An additional interchain hopping parameter t_d couples the chains into a two-dimensional stack.

2 Bandstructure

The eigenvalues of the Hamiltonian are

$$E_{\pm} = \pm\sqrt{\alpha^* + \beta}\sqrt{\alpha + \beta^*}, \tag{5}$$

which are plotted in Figure **??** for the set of parameters $(\mu, v, w, t_d) = (0, 0.6, 1.2, 0.9)$, i.e. the SSH topological phase. In the trivial phase, $(\mu, v, w, t_d) = (0, 1.2, 0.6, 0.9)$, the band structure looks as Figure **??**. At $\mu = 0$, the bands are seen to cross at zero-energy, possibly Dirac node.

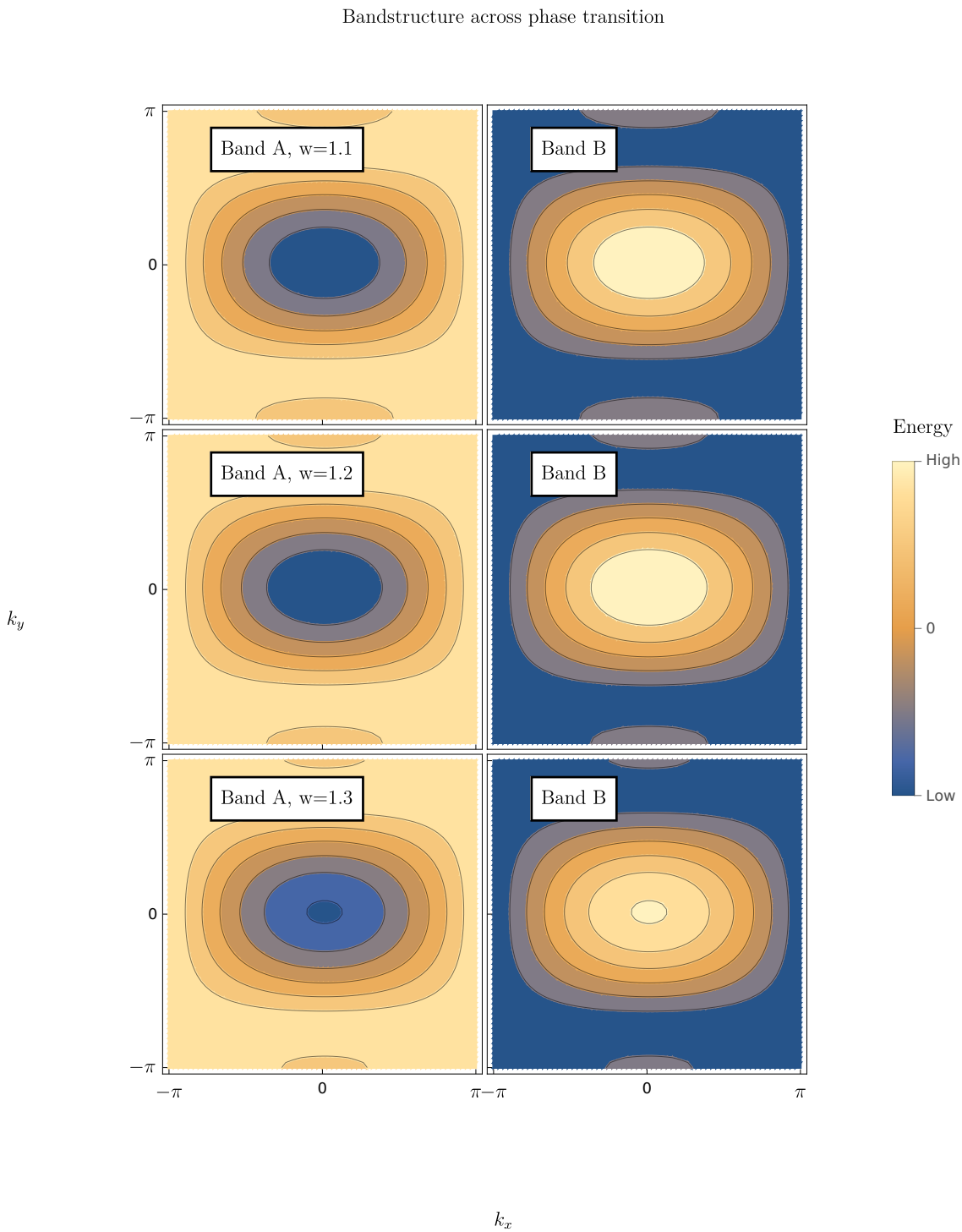

Figure 2: Bandstructure of the NiC_2 planes in $LaNiC_2$ as the SSH dimers cross their topological phase transition, with $(\mu, v, t_d) = (0, 1.2, 0.9)$. At $E = 0$, a band-crossing occurs.

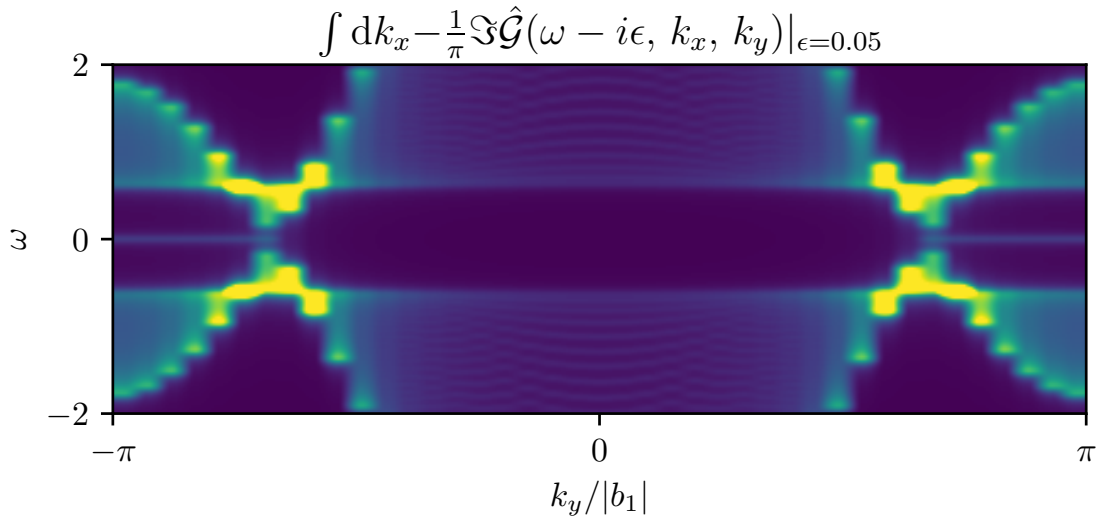

Figure 3: The spectrum of the simple NiC_2 model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The SSH dimers are in their topological state with $\mu = 0.0, v = 1.2, w = 0.6, t_d = 0.9$. Dirac nodes are observed, which pin Majorana zero-modes. Between the Dirac nodes is a line of zero-energy states, known as a Bogoliubov-Fermi arc.

3 Finite size results: ubiquitous topological modes

Performing a finite size calculation with 41×41 sites, with periodic boundary conditions perpendicular to the axis of the SSH dimers, and open-boundary conditions along their axis, we expected to observe topological features, namely Majorana and Bogoliubov-Weyl modes, due to the simple model's simiarities to a recently studied model [CITE: Rosenberg and Manousakis arXiv:2203.12004v1]. However, in addition to similar modes obeserved in both models' topological phases (see Figure 3), in our simple model we surprisingly observe the same topological features in the parameter regime corresponding to the trival phase of the SSH model, see Figure 4.

Crossing the phase transition, we observe no change in the bandstructure Figure 2, so the transition appears to be topological.

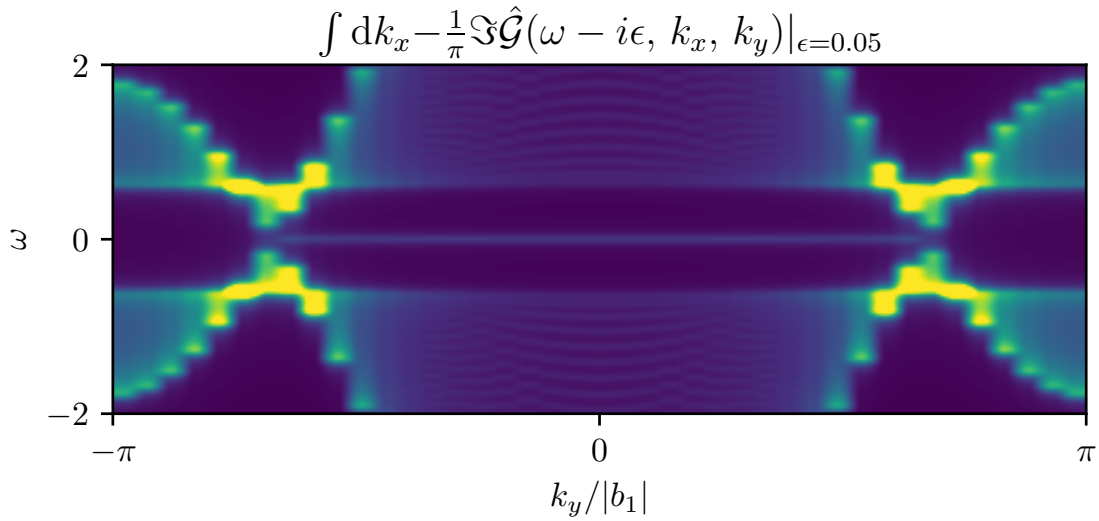

Figure 4: The spectrum of the simple NiC_2 model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The SSH dimers are in their trivial state with $\mu = 0.0, v = 0.6, w = 1.2, t_d = 0.9$. Curiously, Majorana modes remain, as well as the Bogoliubov-Fermi arc, with a $k = \pi$ phase difference.

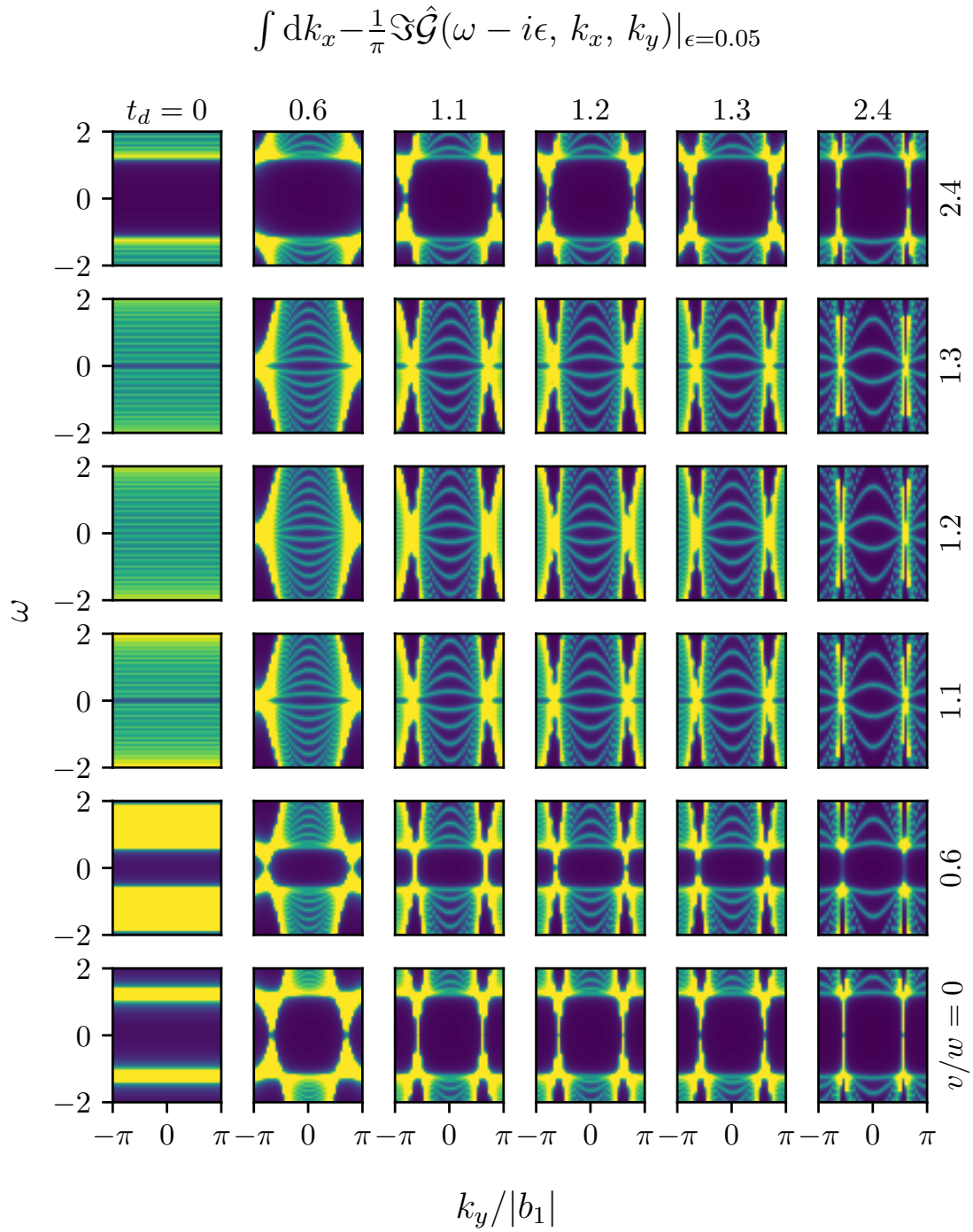

Figure 5: The spectrum of the simple NiC_2 model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The model consists of an 43×43 lattice, with closed boundary conditions. The chemical potential is $\mu = 0.0$, the interdimer hopping is fixed $w = 1.2$, and the intradimer v and interchain t_d hoppings are varied. Compare with the model with open boundary conditions Fig 6.

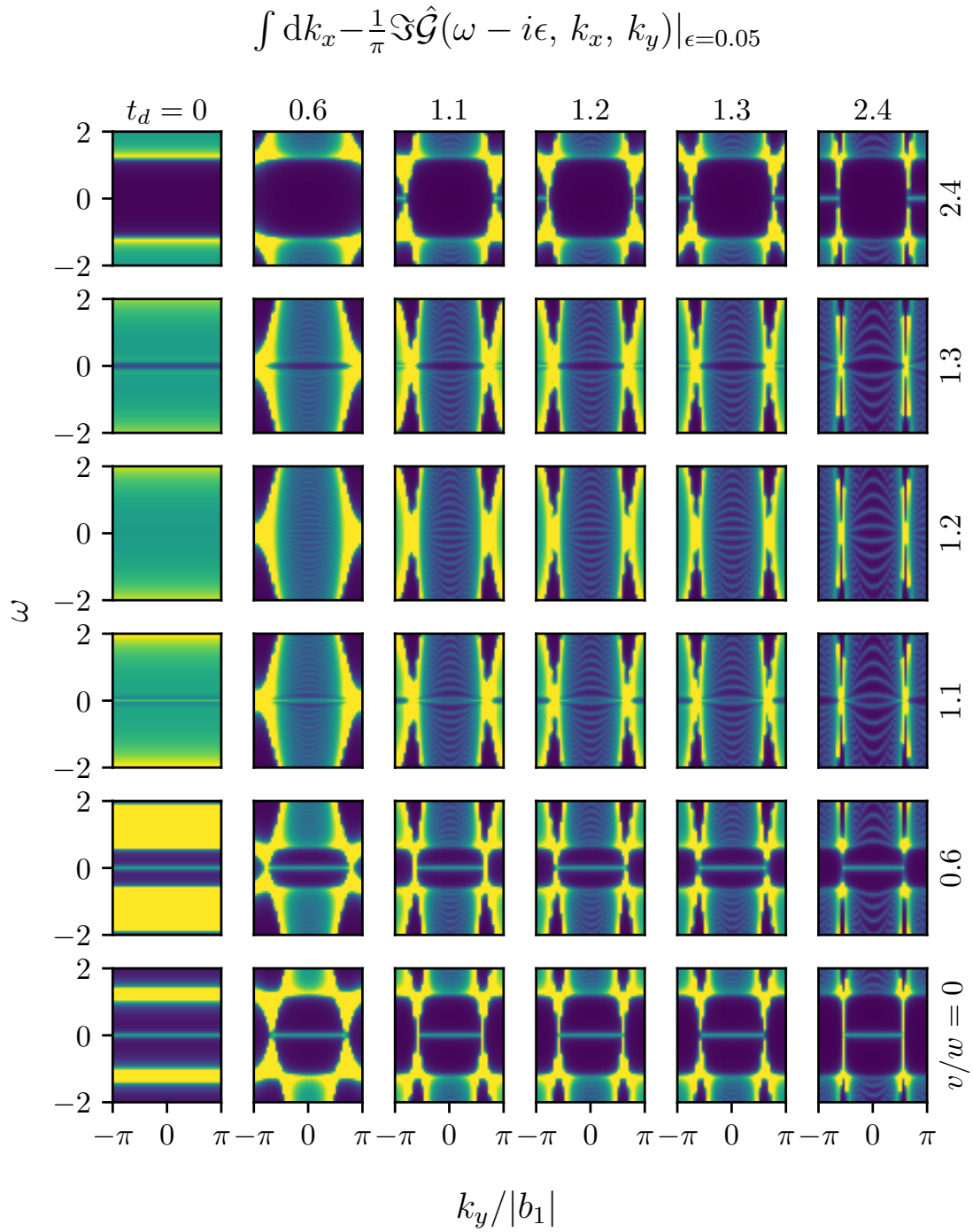

Figure 6: The model and quantities depicted as in Fig 5, but with the boundary conditions closed. We observe the diagonal hopping t_d has the effect of *squeezing* the topological modes into a narrower interval $\mathcal{I} \subset [-\pi, \pi]$. At the conventional SSH topological transition point $v = w$, the narrow interval transitions to its complement interval $\mathcal{I}^c$. Notice the universality with the NiC_2 model Fig **??**, **??**.